MIL-S-22432/30C
22 August 1974

SUPERSEDING
MIL-S-22432/30B
19 MARCH 1968
MIL-S-22432/30A(WP)
21 APRIL 1966
MIL-S-22432/30(WP)
9 MARCH 1964

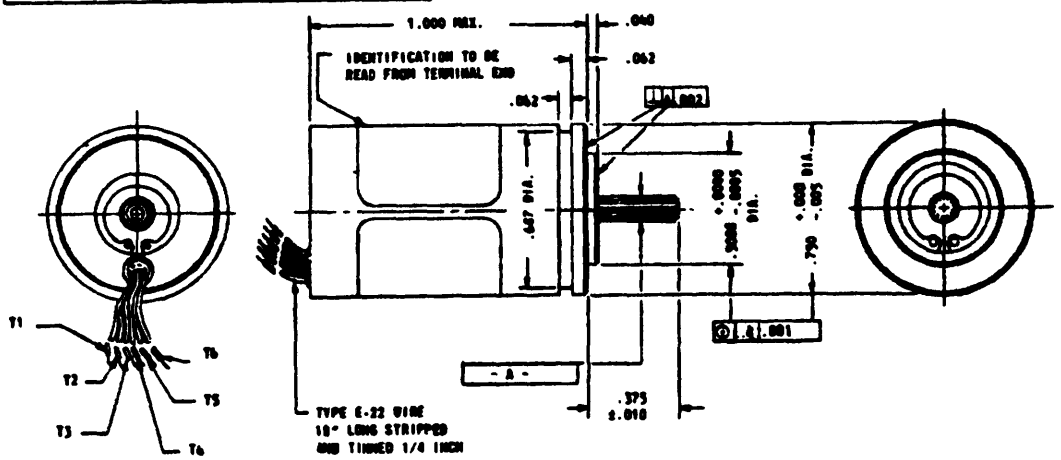
MILITARY SPECIFICATION SHEET
SERVOMOTOR, TYPE 26V-08SM4d

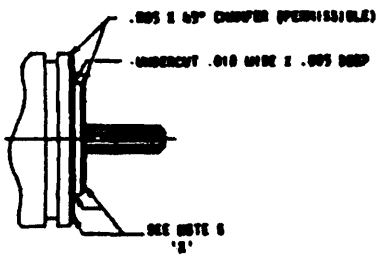
This Specification is Mandatory for Use by All Departments and Agencies of the Department of Defense.

THE COMPLETE REQUIREMENTS FOR PROCURING THE SERVOMOTOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF SPECIFICATION MIL-S-22432

SPUR GEAR DATA	
TEETH	13
PITCH	120
OUTSIDE DIAMETER	.1267 +.0000 -.0010
PRESSURE ANGLE	20°
TOOTH FORM	FULL DEPTH INVOLUTE
DESIGN TO MEET AGMA 207.04 (ASA-B-6.7-1956) "20 DEGREE INVOLUTE FINE PITCH SYSTEM" PRECISION GRADE 1 TESTED TO AGMA 236.04 (ASA-B-6.11-1958) INSPECTION OF FINE GEARS.	

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ON FRACTIONS ±1/64
DECIMALS ±.005 ANGLES ±2°





NOTES:

1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE SHAFT, THEN THE HOUSING SHALL BE ROTATED.
2. RUNOUT OF FEATURE 'A' SHALL NOT EXCEED .0014 TIR.
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING REVERSAL OF THE FOLLOWING LOADS:
   END PLAY - 1/2 POUND RADIAL PLAY - 1/2 POUND
4. TERMINAL LEAD WIRES SHALL DEPART THROUGH THE BACK SURFACE OF THE TERMINAL END OF THE SERVOMOTOR IN ANY ACCEPTABLE MANNER.
5. HOUSING SHALL BE STAINLESS STEEL. SURFACES MARKED 'Z' SHALL BE BRIGHT METALLIC FINISH (AS PER MIL-C-13924).
6. DIMENSIONS AND TOLERANCES COMPLY WITH ANSI Y14.5.
7. PENDING ISSUANCE OF A QUALIFIED PRODUCTS LIST (QPL) A SAMPLE OF THE SERVOMOTORS TO BE FURNISHED UNDER THIS SPECIFICATION SHALL BE SUBJECTED TO THE PREPRODUCTION TESTS AS SPECIFIED IN SPECIFICATION MIL-S-22432.

FSC 6105

FIGURE 1

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TABLE I
SERVOMOTOR, TYPE 26V-08SM4d

Performance Requirements and Characteristics

Requirements	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	hz	
Test	400	hz	±1%
Range	380 to 420	hz	
Voltage:			
Reference phase, rated	26	volts rms	
Reference phase, test	26	volts rms	±1%
Reference phase, range	23.4 to 28.6	volts rms	
Control phase, rated (series)	36	volts rms	
Control phase, test (series)	36	volts rms	±1%
Control phase, range (series)	0 to 40	volts rms	
Control phase, rated (parallel)	18	volts rms	
Control phase, range (parallel)	0 to 20	volts rms	
Current:			
Reference phase	.245	amperes rms	maximum
Control phase (series)	.178	amperes rms	maximum
Control phase (parallel)	.318	amperes rms	nominal
Power:			
Reference phase	3.6	watts	maximum
Control phase (series)	3.6	watts	maximum
Control phase (parallel)	3.4	watts	nominal
Synchronous Speed	8000	rpm	
Impedance:			
Reference phase	67+j95	ohms	nominal
Control phase (series)	122+j171	ohms	nominal
Power Factor:			
Reference phase	.58		nominal
Control phase (series)	.58		nominal
Effective Resistance:			
Reference winding	204	ohms	nominal
Control phase (series)	366	ohms	nominal
Control phase (parallel)	92	ohms	nominal
Minimum Starting Voltage	.75	volts rms	maximum
Coupling in Control Winding	.6	ratio	minimum
Mechanical:			
No-load Speed	6200	rpm	minimum
Stall Torque			
at 23°C	.30	ounce-inches	minimum
at 125°C	.20	ounce-inches	minimum

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TABLE I (Continued)

Requirements	Value	Unit	Tolerance
Rotor Moment of Inertia	.20	gm-cm^2	maximum
Radial Play	.0008	inch	maximum
End Play	.0035	inch	maximum
	.0015	inch	minimum
Temperature Rise	85	degrees C	maximum
Noise, Mechanical	not excessive		
Time Constant	.006	seconds	nominal
Reversing Time	.010	seconds	nominal
Theoretical Acceleration at Stall	106000	radians/second^2	nominal
Maximum Power Output	.4	watts	nominal
Weight	2.0	ounces	maximum

Notes:

1. The servomotor identified herein shall conform to all requirements, and shall be subjected to all tests delineated in MIL-S-22432, with the following exceptions:

Salt Atmosphere Resistance, paragraph 3.30
Audible Noise, paragraph 3.31

2. Type 26V-08SM4d servomotors shall be capable of operation in ambient temperatures ranging from -55°C to 125°C.

3. Connection Diagram:

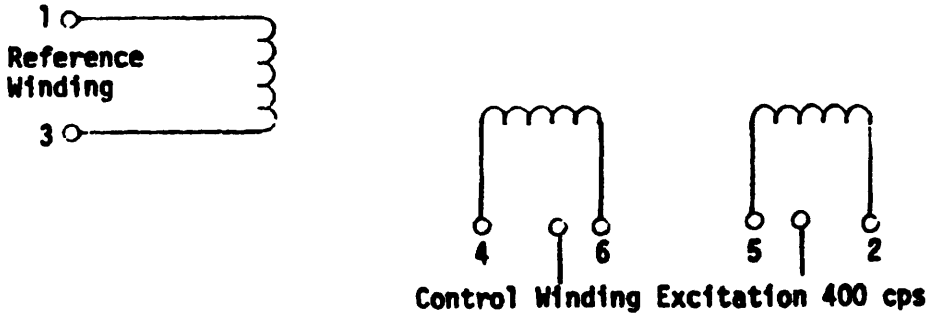


Figure 2

a. For CCW rotation the voltage at terminal 2 leads the voltage at terminal 1 by 90 electrical degrees.
b. For series operation of control winding terminals 5 and 6 are strapped and 36 volts rated is applied across terminals 2 and 4.
c. For multiple or parallel operation of control winding terminals 6 and 2 are strapped, and terminals 4 and 5 are strapped. 18 volts rated is applied across terminals 2 and 4.

4. Sampling, inspection and preparation for delivery shall be in accordance with MIL-S-22432.

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Custodians:

Army - MU
Navy - AS
Air Force - 11

Review Activities:

Army - EL, MI, MU
Navy - SH, AS, OS, EC
Air Force - 11, 82

User Activities:

Army - ME, AV
Navy - OS, MC, CG

Preparing Activity:

Navy - AS

DOD Project Number:

6105-0110

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